PARUL INSTITUTE OF ENGINEERING & TECHNOLOGY

Department of AI & Data Sciences | Subject: Linear Algebra (203105400)

Assignment 3 — Question Paper

Question 1 (5 Marks):
Given the 2×2 real matrix:

$$A = (4)\,123$$

[label=()]

1. Set up the characteristic matrix equation $\det(A - \lambda I) = 0$.
2. Compute the determinant and expand to find the characteristic polynomial.
3. Solve for all eigenvalues λ_1, λ_2 of matrix A. Show all factorization steps.